Exercise Sheet 16

The notation is that of Lecture 16. The invariant form is the course form, $\operatorname{tr}(XY)$ on $\mathfrak{sl}_n$. Verma modules and $L(\mu)$ with $\mu \notin P_+$ are infinite-dimensional. Their highest-weight submodules and quotients may also be infinite-dimensional. Every other module on this sheet is finite-dimensional. Exercises 16.1–16.2 repeat the proof in rank one. Exercises 16.3–16.4 show which step of the proof removes each false candidate. Exercises 16.5–16.6 transfer the method.

Exercise 16.1 (Casimir normalisations in $\mathfrak{sl}_2$). Use the basis (h, e, f) and the form $(X, Y) = \operatorname{tr}(XY)$.

(i) Find the dual basis and show $\Omega = \frac{1}{2}h^2 + ef + fe = \frac{1}{2}h^2 + h + 2fe$.

(ii) Compute the scalar of Ω on $M(z\omega)$, $z \in \mathbb{C}$, and compare it with $(z\omega, (z+2)\omega)$.

(iii) Show that the Killing form is $4\operatorname{tr}(XY)$ and give the Killing Casimir scalar.

(iv) For $m \in \mathbb{Z}_{\geq 0}$, verify directly that Ω acts on $f^{m+1}v_0 \in M(m\omega)$ by $m(m+2)/2$.

Exercise 16.2 (The proof in rank one). Let $\mathfrak{g} = \mathfrak{sl}_2$, $m \in \mathbb{Z}_{\geq 0}$, $\lambda = m\omega$ and $x = e^{\omega}$.

(i) Determine S_λ.

(ii) Using Steps 1–2 of the sphere decomposition, prove that $M(-(m+2)\omega)$ is irreducible.

(iii) Using the singular vector of Lecture 12, write the sphere decomposition of $\operatorname{ch} M(m\omega)$ and invert it.

(iv) Carry out steps (a)–(e) of the proof for $A_\rho \operatorname{ch} L(\lambda)$. Which step is not needed in rank one?

Exercise 16.3 (Weyl invariance does not determine an irreducible character).

(i) For $\mathfrak{sl}_2$, show that $e^{2\omega} + e^{-2\omega}$ is W-invariant, compute $A_\rho(e^{2\omega} + e^{-2\omega})$, and show that it is not the character of any module.

(ii) For $\mathfrak{sl}_3$ and χ_m as in the lecture, find all $m \in \mathbb{Z}_{\geq 0}$ for which χ_m is the character of a module, and those for which it is irreducible.

(iii) Which step of the lecture's proof excludes χ_m, $m \neq 2$, as the character of $L(\theta)$?

Exercise 16.4 (The sphere is not enough). Let $\mathfrak{g} = \mathfrak{sl}_3$ and $\lambda = 3\omega_1$, so $\lambda + \rho = 4\omega_1 + \omega_2$.

(i) Show $(\omega_1, \omega_1) = (\omega_2, \omega_2) = \frac{2}{3}$ and $(\omega_1, \omega_2) = \frac{1}{3}$.

(ii) Find all $\nu \in P$ with $\|\nu\|^2 = \|\lambda + \rho\|^2$ and split them into W-orbits.

(iii) Give an alternating element with coefficient 1 at $\lambda + \rho$ and all exponents on this sphere, other than $A_{\lambda+\rho}$. Which step of the proof excludes it?

Exercise 16.5 (Triangular inversion and the Verma expansion).

(i) Let $(E, \leq)$ be a finite poset enumerated as $\mu_1, \dots, \mu_l$ with $\mu_i < \mu_j \Rightarrow i < j$. Let A be an integer matrix with $a_{kk} = 1$ and $a_{kj} = 0$ unless $\mu_j \leq \mu_k$. Prove that A^{-1} has the same two properties.

(ii) Deduce that in the candidate-weight theorem one may take $b_\mu = 0$ unless $\mu \leq \lambda$.

(iii) Let $\lambda \in P_+$ and $w \cdot \lambda = w(\lambda + \rho) - \rho$. Prove the Verma expansion $\operatorname{ch} L(\lambda) = \sum_{w \in W} \det(w) \operatorname{ch} M(w \cdot \lambda)$.

(iv) For the adjoint module of $\mathfrak{sl}_3$, list $w \cdot \theta$ and use the Verma expansion to recover the multiplicities of 0 and α_1.

Exercise 16.6 (Clebsch–Gordan from the character formula). Let $m, n \in \mathbb{Z}$ with $m \geq n \geq 0$, and let $x = e^{\omega}$ for $\mathfrak{sl}_2$.

(i) Prove that finite-dimensional $\mathfrak{sl}_2$-modules with equal characters are isomorphic.

(ii) Using the Weyl character formula, prove $\operatorname{ch} V_m \operatorname{ch} V_n = \sum_{k=0}^{n} \operatorname{ch} V_{m+n-2k}$.

(iii) Deduce $V_m \otimes V_n \cong \bigoplus_{k=0}^{n} V_{m+n-2k}$ and check $V_2 \otimes V_1$.